Pre-Analysis Plan:
Impact of Having Daughters on Attitudes, Preferences, and Behaviors

#ManyDaughters

Katharina Stückradt* Lennart Schürmann†

1 Study information

1.1 Description

This pre-analysis plan outlines the empirical strategy for examining the relationship between child gender composition and gender-related attitudes and behaviors among parents in Germany. Using data from the German Socio-Economic Panel (SOEP), we test a set of pre-registered hypotheses regarding whether having daughters—as opposed to sons—is associated with stronger support for gender equality, more progressive political preferences, and more equal divisions of household labor.

We leverage high-quality panel data to construct both attitudinal and behavioral outcome measures, and focus on parents whose first child was born before 2005. The plan includes detailed information on variable construction, sample restrictions, model specifications, and controls. By clearly defining our hypotheses and analytical steps in advance, we aim to enhance the transparency and credibility of our research findings.

1.2 Hypotheses

- H1: Having daughters leads to stronger support for gender equality (as measured by attitudes).
- H2: Having daughters leads to more left-leaning political preferences.
- H3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
- H4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

*University of Amsterdam; ✉ k.e.stuckradt@uva.nl
†Freie Universität Berlin & WZB Berlin Social Science Center; ✉ lennart.schuermann@wzb.eu

1.3 Available Data

Transparency explanation of available data: By the time of writing this pre-analysis plan, the authors had already access to a sample of the dataset. We ensure that the full data set could not be opened by any of us. We only tested the availability of key variables in the panel years and data quality with a 5% data sample, which does not engage with actual analysis.

1.4 Data Source

We will use the German Socio-Economic Panel (SOEP) data provided by the Many Daughters project coordinators. The SOEP is a longitudinal household survey that has been conducted annually since 1984.

2 Variables

2.1 Dependent Variables

Attitudes (H1 + H4):

- plj0620: democracy: women have the same rights as men
- plj0621: position of women: working as women
- plj0622: position of women: working as married women
- plj0623: position of women: when woman earns more than partner, problems
- plj0624: position of women: education of son more important than of daughter
- plj0625: position of women: husband should have the final say

To construct a measure of support for gender equality, we use several survey items capturing respondents' attitudes toward women's rights and roles in society. Each item originally ranges from either 0–10 or 1–7. The variable construction proceeded as follows:

Variable Selection and Cleaning: We include five attitudinal items (plj0621 to plj0625; we eventually do not use plj0620 as this is not available given our sample limitations) that assess gender-related beliefs, such as support for equal rights, women's independence through work, and traditional views on gender roles. All responses with invalid or negative values (e.g., "don't know") are recoded as missing.

Rescaling and Harmonization: Three items (plj0623, plj0624, and plj0625) express more traditional, less egalitarian views when scored high. These need to be reverse-coded so that, across all items, higher values consistently reflect greater support for gender equality.

Reliability Testing: We test the internal consistency of the six items using Cronbach's alpha. We only retain the items that align well with each other and produce an acceptable alpha (¿0.7).

Index Construction: A composite score will be created by calculating the mean of the selected items for each respondent. To simplify interpretation, this mean will be rounded and

then recoded into a 5-point ordinal scale, where: 1 = least support for gender equality; 5 = strongest support for gender equality

Aggregation Across Years: To construct a stable, longitudinal measure of gender attitudes, we averaged this 5-point scale for each individual across the survey years 2016 to 2018, where the data is available. This generated a time-aggregated measure of gender equality attitudes per respondent. The averaged variable will also be rounded and recoded into a 5-point scale, and labeled as the final dependent variable $genderatt_1618$.

Political Preferences (H2):

- plh0004: Political attitude

To measure respondents' political orientation, we use the variable plh0004, which captures self-placement on a left–right ideological scale ranging from 0 to 10, where 0 indicates a completely left-leaning position and 10 indicates a completely right-leaning position. To ensure consistency and comparability across hypotheses, we focus on data from the 2019 survey wave, aligning with other dependent variables that are only available for 2016–2018.

Cleaning and Rescaling: We treat all negative values as invalid responses (e.g., nonresponse or "don't know") and recode them as missing. Because we are specifically interested in left-leaning political preferences, we reverse the original scale so that higher values reflect more left-leaning attitudes. This results in a new variable, $revplh0004$, where: 0 represents the most right-leaning, 10 represents the most left-leaning. This transformation makes it easier to interpret regression results: positive coefficients now indicate a stronger association with left-leaning political views.

Gender Equality in Household Labor (H3)

- pli0043_h: hours spent on housework
- pli0044_h: hours spent on childcare

To measure gender equality in household behavior and outcomes, we construct a variable that captures how equally domestic and childcare responsibilities are shared between partners. We base this measure on two survey items: `pli0043_h` (hours spent on housework) and `pli0044_h` (hours spent on childcare).

Data Cleaning: We first recode all negative values in `pli0043_h` and `pli0044_h` as missing, as these likely indicate invalid responses such as "don't know" or refusals.

Merging Partner Data: To compare household labor contributions between partners, we create a temporary dataset containing each respondent's housework and childcare hours, along with their personal ID (`pid`) and survey year (`syear`). We rename `pid` to `parid` to represent the partner ID and rename the variables to reflect that they belong to the partner. This dataset is saved and later merged with the main dataset using both `parid` and `syear`. We also create "self" versions of the housework and childcare variables to distinguish the respondent's values from their partner's after the merge.

Calculating Labor Shares: After merging, we compute:

- the total hours spent on housework and childcare by the **respondent** (`housechild_work_self`),

- the total hours for the **partner** (`housechild_work_partner`),
- and the **combined total** for both partners (`total_housechild`).

We then calculate the key indicators:

- **Equality of Task Distribution:** `housechild_equal` measures how evenly tasks are distributed between partners, calculated as:

$$\texttt{housechild_equal} = 1 - \frac{|\texttt{self hours} - \texttt{partner hours}|}{\texttt{total hours}}$$

  This score ranges from 0 (completely unequal distribution) to 1 (perfect equality).

Creating the Final Dependent Variable: To obtain a stable measure over time, we average `housechild_equality` across the years 2016 to 2018 for each respondent. This produces the variable `housechild_equality_1618`. Finally, we round it to one decimal place and then multiply by 10 to create a categorical version (`housechild_equal_cat`) with 11 possible values, ranging from:

- **0 = no gender equality** (one partner does all the work)
- **10 = perfect gender equality** (tasks are shared equally).

This final variable, `housechild_equal_cat`, serves as the dependent variable for Hypothesis 3 (H3) on gender equality in household labor distribution.

2.2 Independent Variable

Having a Daughter:

- kidsex01: Sex Child 1 - kidsex18: Sex Child 18

To capture whether a respondent has at least one daughter, we construct a binary (dummy) variable named `daughter`. Respondents without any children are excluded from the sample and therefore not considered in this construction. The dataset includes variables `kidsex01` through `kidsex18`, which record the sex of up to 18 children for each respondent. For each of these, we check whether the value is equal to 2 (indicating a female child). If at least one of them is coded as female, the `daughter` variable is set to 1.

We define the variable as follows:

- `daughter = 1` if the respondent has at least one daughter.
- `daughter = 0` if the respondent has children, but only sons.

2.3 Control Variables

In addition to the main independent variables, we include several control variables that may influence the dependent variables. Below is an overview of how each control variable is constructed and used:

Sisters

We generate a dummy variable `sister` coded as 1 if the respondent has at least one sister, based on the sex of siblings recorded in variables `pla0017`, `pla0020`, ..., `pla0044`. A value of 2 indicates a female sibling. If at least one sibling is female, `sister` is set to 1; otherwise, it remains 0.

Year of Birth

The respondent's year of birth (`gebjahr`) is included as a categorical variable using factor notation (`i.gebjahr`) to control for cohort effects.

Personal Net Income

Net personal income is measured using the variable `pglabnet`. Negative values, which likely indicate invalid or missing responses, are recoded as missing in a cleaned version called `pglabnet_mi`. Because income is typically right-skewed, we apply a natural logarithmic transformation to create the variable `log_netincome`, which is used in the models as a continuous variable (`c.log_netincome`).

Education

Education level is measured with the ISCED-97 classification (`pgisced97`). A cleaned version, `pgisced97_mi`, recodes invalid values (e.g., `-1`) as missing. The variable is treated as categorical in the analysis using factor notation (`i.pgisced97_mi`).

Marital Status

Marital status in the survey year is captured by the variable `pgfamstd`. A recoded version, `pgfamstd_mi`, replaces invalid responses (e.g., `-1`, `-2`, `-3`) with missing values. This variable is included in the analysis as a categorical variable (`i.pgfamstd_mi`).

Migration Background

Migration background is measured using the variable `migback`. This variable is included in the analysis as a categorical variable (`i.migback_mi`).

Gender

Gender is measured using the variable `male`. This variable is a recoded version of the categorical variable `sex` (1 = male, 2 = female). We created the dummy variable `male`, which is 1 for male.

Note: We might omit control variables from the analysis in case the total N becomes too small.

3 Analysis Plan

3.1 Data exclusion

Sample Selection

The analytical sample is restricted to respondents who meet the following criteria:

- Private households only: We retain only respondents who live in private households, as indicated by the variable `pop` taking values 1 or 2. Observations from non-private households are excluded.
- Respondents with children: We exclude all respondents who report having no children, based on the variable `sumkids`. Only respondents with one or more children are included in the analysis.
- First child born before 2005: We restrict the sample to respondents whose first child `kidgeb01` was born before 2005 to ensure sufficient time for daughter effects to manifest. Theoretical literature suggests parents become increasingly aware of gender-specific challenges their daughters face during adolescence, particularly around dating, educational choices, and career planning. With outcome measures in 2016-2018, this ensures children are at least 11-13 years old—an age when gender role socialization intensifies and parents may adjust their attitudes and behaviors in response to observing their daughters navigate gender-related challenges. This cutoff also ensures that children are old enough for long-term patterns of gender role behavior (e.g., household labor division) to have had time to develop.
- Respondent in heterosexual relationship: We restrict the sample to respondents who are in a heterosexual relationship. The literature suggests that individuals in homosexual relationships differ in their gender attitudes from individuals in heterosexual relationships. This is necessary to test hypothesis 4, as we can assess gender equality in distribution of household tasks only for heterosexual partners.

After applying these filters, the sample consists exclusively of parents in private households whose parenting experience predates 2005.

3.2 Statistical Models

Hypothesis Test for H1: Effect of Having Daughters on Gender Equality Attitudes

To test Hypothesis 1, we examine whether having at least one daughter is associated with stronger support for gender equality in attitudes. We estimate an Ordinary Least Squares (OLS) regression model where the dependent variable is `genderatt_1618`, a composite measure of gender-equal attitudes ranging from 1 to 5.

The main independent variable is a binary indicator `daughter`, which equals 1 if the respondent has at least one daughter and 0 if the respondent has sons only.

The attitude variables are only available from 2016 until 2018. We restrict the sample to the years 2016 to 2018, consistent with the construction of the dependent variable. Robust standard errors are used to correct for potential heteroskedasticity.

Model Specification. The regression model is specified as:

$$\begin{aligned}\text{genderatt}_i = \beta_0 &+ \beta_1 \cdot \text{daughter}_i + \beta_2 \cdot \text{sister}_i + \beta_3 \cdot \log(\text{netincome}_i) \\ &+ \beta_4 \cdot \text{education}_i + \beta_5 \cdot \text{maritalstatus}_i + \beta_6 \cdot \text{birthyear}_i \\ &+ \beta_7 \cdot \text{migrationbackground}_i + \beta_8 \cdot \text{male}_i + \varepsilon_i\end{aligned}$$

Where:

- genderatt_i is the dependent variable measuring gender-equal attitudes,
- daughter_i is a binary indicator for having at least one daughter,
- sister_i Dummy variable equal to 1 if the respondent has at least one sister, 0 otherwise.
- $\log(\text{income}_i)$ is the log of net personal income,
- education_i is a categorical variable for education,
- maritalstatus_i is a variable for marital status,
- birthyear_i is a categorical variable for birth cohort,
- $\text{migrationbackground}_i$ is a dummy variable for migration background,
- male_i is a dummy variable equal to 1 if the respondent is male, 0 if female,
- ε_i is the error term.

Estimation We estimate this model using robust standard errors to account for heteroskedasticity:

Hypothesis Test for H2: Effect of Having Daughters on Left-Leaning Political Preferences

To test Hypothesis 2, we investigate whether having at least one daughter is associated with more left-leaning political preferences. We estimate an Ordinary Least Squares (OLS) regression model where the dependent variable is `revplh0004`, a reverse-coded measure of political orientation ranging from right (low values) to left (high values). The original scale, `plh0004`, ranges from 0 (completely left) to 10 (completely right), and is reverse-coded to ensure interpretability consistent with our hypothesis.

The main independent variable is a binary indicator `daughter`, which equals 1 if the respondent has at least one daughter and 0 if the respondent has sons only.

The political preference variable is only available for the year 2019, so the analysis is restricted to that survey year. Robust standard errors are used to account for potential heteroskedasticity.

Model Specification. The regression model is specified as:

$$\begin{aligned}\text{leftleaning}_i = \beta_0 &+ \beta_1 \cdot \text{daughter}_i + \beta_2 \cdot \text{sister}_i + \beta_3 \cdot \log(\text{netincome}_i) \\ &+ \beta_4 \cdot \text{education}_i + \beta_5 \cdot \text{maritalstatus}_i + \beta_6 \cdot \text{birthyear}_i \\ &+ \beta_7 \cdot \text{migrationbackground}_i + \beta_8 \cdot \text{male}_i + \varepsilon_i\end{aligned}$$

Where:

- leftleaning_i is the reverse-coded political attitude variable (higher = more left-leaning),
- daughter_i is a binary indicator for having at least one daughter,
- sister_i Dummy variable equal to 1 if the respondent has at least one sister, 0 otherwise.
- $\log(\text{netincome}_i)$ is the log of net personal income,
- education_i is a categorical variable for education (ISCED-97),
- maritalstatus_i is a variable for marital status,
- birthyear_i is a categorical variable for birth cohort,
- $\text{migrationbackground}_i$ is a dummy variable for migration background,
- male_i is a dummy variable equal to 1 if the respondent is male, 0 if female,
- ε_i is the error term.

Estimation We estimate this model using robust standard errors to account for heteroskedasticity:

Hypothesis Test for H3: Effect of Having Daughters on Gender Equality in Household Behavior

To test Hypothesis 3, we assess whether having at least one daughter is associated with greater gender equality in the division of household and childcare labor. The dependent variable is `housechild_equal`, which ranges from 0 (no gender equality in time spent on housework and childcare) to 10 (perfect equality). It is a discretized version of a continuous measure that captures the balance in hours spent on these tasks between partners.

The main independent variable is `daughter`, a binary indicator coded 1 if the respondent has at least one daughter, and 0 otherwise.

We restrict the analysis to the years 2016 to 2018, in line with the construction of the dependent variable. Because the outcome is based on household-level behavior, we cluster standard errors at the household level (`hid`) to account for potential within-household correlation in the error terms.

Model Specification. The regression model is specified as:

$$\begin{aligned}\text{housechild_equal}_i = \beta_0 &+ \beta_1 \cdot \text{daughter}_i + \beta_2 \cdot \text{sister}_i + \beta_3 \cdot \log(\text{netincome}_i) \\ &+ \beta_4 \cdot \text{education}_i + \beta_5 \cdot \text{maritalstatus}_i + \beta_6 \cdot \text{birthyear}_i \\ &+ \beta_7 \cdot \text{migrationbackground}_i + \beta_8 \cdot \text{male}_i + \varepsilon_i\end{aligned}$$

Where:

- $\text{housechild_equal}_i$ is the dependent variable measuring gender equality in the division of household and childcare labor (0–10),
- daughter_i is a binary indicator for having at least one daughter,
- sister_i Dummy variable equal to 1 if the respondent has at least one sister, 0 otherwise.
- $\log(\text{netincome}_i)$ is the log of net personal income,
- education_i is a categorical variable for education level,
- maritalstatus_i is a categorical variable for marital status,
- birthyear_i is a categorical variable for birth cohort,
- $\text{migrationbackground}_i$ is a dummy variable for migration background,
- male_i is a dummy variable equal to 1 if the respondent is male, 0 if female,
- ε_i is the error term.

Estimation We estimate this model with standard errors clustered at the household level.

Hypothesis Test for H4: Gendered Effect of Having Daughters on Gender Equality Attitudes

Hypothesis 4 investigates whether the effect of having at least one daughter on support for gender equality in attitudes differs by respondent gender. Specifically, we expect that men who have daughters will express more gender-equal attitudes than men without daughters, while the effect for women may be smaller or absent. The dependent variable is `genderatt_1618`, a composite measure of gender-equal attitudes ranging from 1 to 5.

The key independent variable is the interaction between `daughter`, a binary indicator for having at least one daughter, and `male`, a binary indicator equal to 1 for male respondents. This allows us to assess whether the relationship between having daughters and gender attitudes differs by respondent gender.

As in previous models, we restrict the sample to the years 2016–2018, when the attitudinal data are available. We estimate an Ordinary Least Squares (OLS) model with robust standard errors to correct for potential heteroskedasticity.

Model Specification. The regression model is specified as:

$$\begin{aligned}\text{genderatt_1618}_i = \beta_0 &+ \beta_1 \cdot \text{daughter}_i + \beta_2 \cdot \text{male}_i + \beta_3 \cdot (\text{daughter}_i \cdot \text{male}_i) \\ &+ \beta_4 \cdot \text{sister}_i + \beta_5 \cdot \log(\text{netincome}_i) + \beta_6 \cdot \text{education}_i + \beta_7 \cdot \text{maritalstatus}_i \\ &+ \beta_8 \cdot \text{birthyear}_i + \beta_9 \cdot \text{migrationbackground}_i + \varepsilon_i\end{aligned}$$

Where:

- genderatt_1618_i is the dependent variable measuring gender-equal attitudes,
- daughter_i is a binary indicator for having at least one daughter,
- male_i is a dummy variable equal to 1 if the respondent is male, 0 if female,
- $\text{daughter}_i \cdot \text{male}_i$ is the interaction term,
- sister_i Dummy variable equal to 1 if the respondent has at least one sister, 0 otherwise.
- $\log(\text{netincome}_i)$ is the log of net personal income,
- education_i is a categorical variable for education level,
- maritalstatus_i is a categorical variable for marital status,
- birthyear_i is a categorical variable for birth cohort,
- ε_i is the error term.

Estimation We estimate this model using robust standard errors to account for heteroskedasticity:

3.3 Missing data

Respondents with missing data on any of the variables included in a given model are excluded from the analysis. This listwise deletion approach ensures consistency within each regression but may lead to variation in sample sizes across models depending on the availability of the respective variables.

4 Interpretation

4.1 H1: Effect of Having Daughters on Gender Equality Attitudes

To evaluate Hypothesis 1, we examine whether having at least one daughter is associated with more gender-egalitarian attitudes. The key explanatory variable in this analysis is a binary indicator, `daughter`, which takes the value 1 for respondents who report having one or more daughters and 0 for those with sons only. Individuals without children are excluded from the sample, allowing for a focused comparison among parents.

The outcome variable, `genderatt_1618`, represents a continuous index of gender attitudes, based on responses to multiple survey questions administered between 2016 and 2018. Items were recoded where necessary so that higher values consistently reflect stronger support for gender equality.

We assess support for the hypothesis by considering the sign and significance of the regression coefficient on `daughter`. A positive and statistically significant estimate ($p < 0.05$) would indicate that parents with daughters exhibit more egalitarian attitudes than those with only sons. Conversely, a non-significant result suggests that the gender composition of children does not meaningfully shape attitudes toward gender equality in this context.

4.2 H2: Effect of Having Daughters on Left-Leaning Political Preferences

To test Hypothesis 2, we investigate whether having at least one daughter is associated with more left-leaning political preferences. The dependent variable is `revplh0004`, a reversed version of the original political ideology scale ranging from 0 (very left) to 10 (very right). The reversal ensures that higher values correspond to more left-leaning positions, making interpretation consistent with our hypothesis.

The main independent variable is the binary indicator `daughter`, which equals 1 for respondents with at least one daughter and 0 for those with sons only. The analysis is restricted to the year 2019 to maintain comparability with other models where attitudinal measures are available only in earlier years.

We interpret support for the hypothesis based on the coefficient of `daughter` in the regression model. A positive and statistically significant coefficient ($p < 0.05$) would imply that parents with daughters tend to hold more left-leaning political attitudes than those with sons only. If the coefficient is not significant, this would indicate that the presence of daughters does not systematically relate to political orientation within this sample.

4.3 H3: Effect of Having Daughters on Gender Equality in Household Behavior

To evaluate Hypothesis 3, we examine whether having at least one daughter is associated with greater gender equality in the division of household and childcare tasks. The dependent variable is `housechild_equal_cat`, a categorical measure ranging from 0 to 10, where higher values reflect more equal time allocation between partners. It is based on the absolute difference in self- and partner-reported hours spent on household and childcare tasks, normalized by the total time reported by both partners.

The primary independent variable is a binary indicator `daughter`, coded as 1 if the respondent has at least one daughter and 0 if they have sons only. The analysis is restricted to the years 2016–2018, which corresponds to the period during which detailed household labor data are available.

We interpret the results by focusing on the coefficient for `daughter`. A positive and statistically significant coefficient ($p < 0.05$) would suggest that parents of daughters tend to

share household and childcare responsibilities more equally compared to parents of sons only. Conversely, a non-significant result would indicate no systematic relationship between child gender and within-household gender equality in task sharing.

4.4 H4: Gendered Effect of Having Daughters on Gender Equality Attitudes

To assess Hypothesis 4, we investigate whether the association between having at least one daughter and support for gender equality in attitudes differs by the respondent's gender. The dependent variable is `genderatt_1618`, a continuous index (ranging from 1 to 5) that captures gender-egalitarian views based on responses to five attitudinal items, averaged across survey years 2016 to 2018.

The key explanatory variable is the interaction term `daughter * male`, which allows us to test whether the effect of having a daughter varies between male and female respondents. A binary variable `daughter` equals 1 if the respondent has at least one daughter (and 0 otherwise), and `male` is coded as 1 for male respondents and 0 for female respondents.

We interpret the results by examining both the main effects and the interaction coefficient: A statistically significant and positive interaction term ($p < 0.05$) would indicate that the presence of daughters is associated with stronger support for gender equality especially among male respondents. If the interaction is not statistically significant, it suggests that the effect of having a daughter on attitudes toward gender equality does not differ meaningfully between men and women.

5 Limitations

A major limitation of this study is that we are constrained to a cross-sectional analytical approach due to data availability issues. Our dependent variables, particularly the attitude measures (plj0620–plj0625), are only available for a limited number of survey waves in the SOEP dataset, preventing us from implementing a truly longitudinal design that would track within-respondent variation over time.

Ideally, we would observe the same individuals before and after having daughters, allowing us to estimate the causal effect of child gender composition using individual fixed effects. Such a design would control for all time-invariant unobserved characteristics that might correlate with both the likelihood of having daughters and gender-related attitudes or behaviors. However, the restricted availability of our key outcome measures makes this approach infeasible.

As a result of these data constraints, our analysis is fundamentally correlational in nature. While our hypotheses use causal language ("having daughters leads to..."), our cross-sectional design can only identify associations between having daughters and our outcomes of interest. We cannot rule out several alternative explanations for any observed relationships:

- Reverse causality: Pre-existing parental attitudes might influence how they report or perceive their household dynamics and political preferences.
- Omitted variable bias: Unobserved characteristics such as personality traits, family values, or regional culture could simultaneously influence both child gender composition and our dependent variables.

These limitations have important implications for interpreting our results. Any significant associations we find between having daughters and gender-related attitudes, political preferences, or household behaviors should be interpreted as suggestive evidence rather than definitive causal effects. The magnitude and direction of these associations can inform future research and theory development, but cannot be used to make strong claims about the causal impact of child gender on parental outcomes.